Algebraic Geometry Problems

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September 2026

1 Section 1.1

1.1.A Group is a groupoid

(a) Make sense of: "A group is a groupoid with one object".

Answer.

Note that morphism composition is associative, that $id_G \in \mathrm{Mor}(G, G)$, and that $\mathrm{Mor}(G, G)$ is closed. Since G, (as a category) is a groupoid, every morphism in $\mathrm{Mor}(G, G)$ is an isomorphism. Therefore for every $\phi \in \mathrm{Mor}(G, G)$, $\exists! \phi^{-1} \in \mathrm{Mor}(G, G)$, so that

$$\phi \circ \phi^{-1} = \phi^{-1} \circ \phi = id_G$$

Then the set $\mathrm{Mor}(G, G)$ with the operation

$$\circ : \mathrm{Mor}(G, G) \times \mathrm{Mor}(G, G) \to \mathrm{Mor}(G, G)$$

admits group structure. (Dropping the condition that G is a groupoid gives $\mathrm{Mor}(G, G)$ the structure of a monoid.)

(b) Describe a groupoid that is not a group.

Answer.

Let $\mathcal{C}$ be a groupoid with two objects A, B. Note that in a group, composition of any two elements is defined, however in our category $\mathcal{C}$, we have things like $\phi, \psi \in \mathrm{Mor}(A, B)$ which you can not compose because the sources and targets do not match.

1.1.B Automorphism Group

Let $A \in \text{Obj}(\mathcal{C})$. Show that the invertible elements of $\text{Mor}(A, A)$ form a group. (Called the automorphism group $\text{Aut}(A)$). What are the automorphism groups for **Sets** and $\mathbf{Vect}_k$? Prove that isomorphic objects have isomorphic automorphism groups.

Answer.

Let $A \in \text{Obj}(\mathcal{C})$ and consider the subset $\text{Aut}(A) \subseteq \text{Mor}(A, A)$ with the property $f \in \text{Aut}(A) \iff f$ is invertible. Then notes $id_A \in \text{Aut}(A)$, which serves as the identity of the group. As before, morphism composition is associative, and $\forall f \in \text{Aut}(A)$, since f is invertible, $\exists f^{-1} \in \text{Aut}(A)$ such that $f \circ f^{-1} = f^{-1} \circ f = id_A$. Therefore, $\text{Aut}(A)$ satisfies the group axioms.

If $X \in \text{Obj}(\mathbf{Sets})$, $\text{Aut}(X)$ consists of permutations of the elements of X. These are functions $f : X \to X$ which are invertible (bijections). If $V \in \text{Obj}(\mathbf{Vect}_k)$ is a vector space, $Aut(V)$ consists of invertible matrices with entries from k.

Proof. Let A and B be isomorphic objects, so $\exists f : A \to B$ and $g : B \to A$ such that $g \circ f = id_A$ and $f \circ g = id_B$. We know $\text{Aut}(A)$ and $\text{Aut}(B)$ are groups. Define the maps

$$\pi : \text{Aut}(A) \to \text{Aut}(B)$$

by

$$\pi : \phi \mapsto f \circ \phi$$

and

$$\pi^{-1} : \text{Aut}(B) \to \text{Aut}(A)$$
$$\pi^{-1} : \psi \mapsto g \circ \psi$$

and then

$$\pi^{-1}(\pi(\phi)) = g \circ (f \circ \phi) = (g \circ f) \circ \phi = id_A \circ \phi = \phi$$

$$\pi(\pi^{-1}(\psi)) = f \circ (g \circ \psi) = (f \circ g) \circ \psi = id_B \circ \psi = \psi$$

So $\pi^{-1} \circ \pi = id_{\text{Aut}(A)}$ and $\pi \circ \pi^{-1} = id_{\text{Aut}(B)}$. Therefore $\text{Aut}(A) \cong \text{Aut}(B)$. $\square$

2 Section 1.2

1.2.A Initials and Terminals

Definition 1. *An object in a category $\mathcal{C}$ is initial if it has exactly 1 map to each object in $\mathcal{C}$. An object is terminal (or final) if it has exactly one map* from *each object. An object is a zero object if it is both initial and terminal.*

Show that any two initial objects are uniquely isomorphic. Same for terminal.

Proof. Let A, A' be two initial objects. Then, since A is initial, $\exists! f : A \to A'$. Similarly, since A' is initial, $\exists! g : A' \to A$. Then $f \circ g : A \to A$ is unique, but $id_A : A \to A$ is a map $A \to A$. So $f \circ g = id_A$. Similarly, $g \circ f : A' \to A'$ is unique, so $g \circ f = id_{A'}$. Therefore, $A \cong A'$ uniquely. For terminal, flip the arrows. □

1.2.B Examples of initals/terminals/zeros

What are initial, terminal, and zero objects in **Sets**, **Rings**, **Top**?

Answer.

- In **Sets**, the empty set $\emptyset$ maps uniquely into every set X, via the empty function (which acts as the inclusion $\emptyset \hookrightarrow X$), so it is inital. However, you can not define a function $\phi : X \to \emptyset$ for all X, so it is not a terminal object. For any singleton set $\{a\}$, there is a unique map $g : X \to \{a\},\ g(x) = a$ for all $x \in X$. Therefore, each singleton set is terminal.
- In **Rings**, the zero ring $\{0\}$ is terminal, because for any ring R, take $f : R \to \{0\}$, $f(r) = 0$ for all $r \in R$. This is a valid ring hom, and it is unique. The ring $(\mathbb{Z}, +, \cdot)$ is initial, because for any ring R, we must have (by our conventions)

  $$f : \mathbb{Z} \to R,\ f : 1_{\mathbb{Z}} \mapsto 1_R$$

  and since $\mathbb{Z}$ is generated by $1_{\mathbb{Z}}$, that rule determines the entire homomorphism. (Note: this map captures the Characterstic of the ring R, because it is the n in $\ker f = n\mathbb{Z}$).

- In **Top**, similar to **Sets**, the empty space $\emptyset$ (with its corresponding toplogy) is initial via the empty function, which is trivially continuous. The terminal objects are singletons $\{a\}$, because there is a unique map $f : X \to \{a\}$, $f(x) = a$, which is continuous because the preimage of a is X, which is open.

1.2.C Mapping a ring into its localiztion

Show that $A \to S^{-1}A$, $(a \mapsto \frac{a}{1})$ is injective iff S contains no zerodivisors.

Proof. $(\implies)$: Let $f : A \to S^{-1}A$, $f : a \mapsto \frac{a}{1}$ be injective. Let $a \neq b$, then $f(a) \neq f(b) \implies \frac{a}{1} \neq \frac{b}{1}$. That means

$$\nexists\, s \in S, \text{ so that } s(a-b) = 0$$

So for $x = (a-b) \in A$ nonzero, there is no $s \in S$ so that $s \cdot x = 0$. Therefore, S contains no zerodivisors.

$(\impliedby)$: Let S be a multiplicative set with no zerodivisors. Let $f : A \to S^{-1}A$, $f : a \mapsto \frac{a}{1}$ be a function. Let $f(a) = f(b)$ for some $a, b \in A$. Then $\frac{a}{1} = \frac{b}{1} \implies \exists s \in S$ such that $s(a-b) = 0$. Since s is not a zerodivisor, we must have $a - b = 0 \implies a = b$. Therefore f is injective. □

1.2.D Univseral Property of Localization

Verify $A \to S^{-1}A$ satisfies the following universal property: $S^{-1}A$ is initial among A-algebras B where every element of S is sent to an invertible element in B. Translation: any map $A \to B$ where every $s \in S$ is sent to an invertible in B must factor uniquely through $A \to S^{-1}A$.

Remark. *An A-module M is the same as a map*

$$\rho : A \to \mathrm{End}_{\mathbb{Z}}(M)$$

$$\rho : s \mapsto \mu_s$$

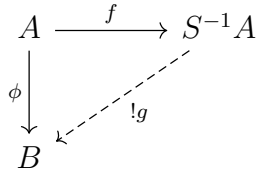
where

$$\mu_s : M \to M$$

$$\mu_s : m \mapsto s \cdot m$$

And to say that a scalar s is invertible is to say that $\mu_s \in \mathrm{End}_{\mathbb{Z}}(M)$ is a bijection (automorphism).

Proof. Let $f : A \to S^{-1}A$ where $f(a) = \frac{a}{1}$. Let B be an A-algebra, with map $\phi : A \to B$ such that $\phi(s)$ is a unit in B. We claim $\exists! g$ such that



commutes. Uniqueness is shown by the following:

$$g\left(\frac{a}{s}\right) = g(f(a) \cdot f(s)^{-1}) = g(f(a)) \cdot g(f(s))^{-1} = \phi(a) \cdot \phi(s)^{-1}$$

So what g does to $\frac{a}{s} \in S^{-1}A$ is completely determined by $\phi : A \to B$. To show existence we need to show that g is well-defined and that it is a ring hom.

Well-defined: Let $\frac{a'}{s'} = \frac{a}{s}$. Then we want $g\left(\frac{a'}{s'}\right) = g\left(\frac{a}{s}\right)$.

$$\frac{a'}{s'} = \frac{a}{s} \implies \exists t \in S \text{ such that } t(a's - s'a) = 0$$

This is an expression in A, so apply $\phi : A \to B$ to both sides:

$$\phi(t)(\phi(a's) - \phi(as')) = 0$$

Since $t \in S$, $\phi(t)$ is a unit, so it is not a zerodivisor. Therefore $\phi(a's) = \phi(as') \implies \phi(a')\phi(s) = \phi(a)\phi(s')$. Since $\phi(s)$ and $\phi(s')$ are units in B, we can multiply by $\phi(s)^{-1}$ and $\phi(s')^{-1}$ to get:

$$\phi(a')\phi(s')^{-1} = \phi(a)\phi(s)^{-1} \implies g\left(\frac{a'}{s'}\right) = g\left(\frac{a}{s}\right)$$

As required, so g is well-defined.

Ring homomorphism: We check $g(1_{S^{-1}A}) = 1_B$:

$$\begin{aligned} g\left(\frac{1}{1}\right) &= \phi(1_A)\phi(1_A)^{-1} \\ &= 1_B \cdot 1_B \\ &= 1_B \end{aligned}$$

g respects addition:

$$\begin{aligned} g\left(\frac{a}{s}+\frac{b}{t}\right) &= g\left(\frac{at+bs}{st}\right) \\ &= \phi(at+bs)\phi(st)^{-1} \\ &= \left[\phi(a)\phi(t)+\phi(b)\phi(s)\right]\phi(s)^{-1}\phi(t)^{-1} \\ &= \phi(a)\phi(s)^{-1}+\phi(b)\phi(t)^{-1} \\ &= g\left(\frac{a}{s}\right)+g\left(\frac{b}{t}\right) \end{aligned}$$

g respects multiplication:

$$\begin{aligned} g\left(\frac{a}{s}\cdot\frac{b}{t}\right) &= g\left(\frac{ab}{st}\right) \\ &= \phi(ab)\phi(st)^{-1} \\ &= \phi(a)\phi(b)\phi(s)^{-1}\phi(t)^{-1} \\ &= \phi(a)\phi(s)^{-1}\phi(b)\phi(t)^{-1} \\ &= g\left(\frac{a}{s}\right)\cdot g\left(\frac{b}{t}\right) \end{aligned}$$

So

$$\begin{aligned} g &: S^{-1}A \to B \\ g &: \frac{a}{s} \mapsto \phi(a)\phi(s)^{-1} \end{aligned}$$

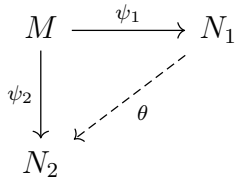
is a well defined, unique ring homomorphism which makes the diagram commute. Therefore, every map $\phi : A \to B$ (every A-algebra B) factors uniquely through $S^{-1}A$. □

1.2.D.1 Localization of modules, mini problem:

Let M be an A-module. Then we define $\phi : M \to S^{-1}M$ as being initial among A-module maps $M \to N$ such that elements of S are invertible in N. Or, $M \to S^{-1}$ is universal among A-module maps from M to modules that are actually $S^{-1}A$-modules. Mini problem: Define the objects and morphisms in this category to make this precise.

Answer.

- The objects of this category are pairs (N_i, ψ_i) where N_i is an A-module and $\psi : M \to N_i$ is an A-module homomorphism such that the multiplication map $s \times - : N_i \to N_i$ is an automorphism.
- The morphisms of this category are: Let (N_1, ψ_1), (N_2, ψ_2) be two objects of the category. Then they both define maps from M:



  The map $\theta : N_1 \to N_2$ of A-modules, such that the triangle commutes $(\theta \circ \psi_1 = \psi_2)$ is an element of $\mathrm{Mor}((N_1, \psi_1), (N_2, \psi_2))$.

1.2.E Localization of Modules

Show that $\phi : M \to S^{-1}M$ exists by constructing something satisfying the universal property. Hint: define elements of $S^{-1}M$ to be $\frac{m}{s}$ where $m \in M$, $s \in S$, with equality given by

$$\frac{m_1}{s_1} = \frac{m_2}{s_2} \iff \exists t \in S \mid t(m_1 s_2 - m_2 s_1) = 0$$

Addition given by

$$\frac{m_1}{s_1} + \frac{m_2}{s_2} = \frac{m_1 s_2 + m_2 s_1}{s_1 s_2}$$

And the $S^{-1}A$-module structure (and therefore the A-module structure) is given by

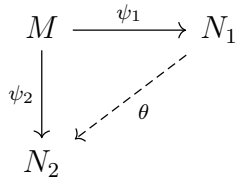
$$\frac{a}{s} \cdot \frac{m_1}{t} = \frac{a \cdot m_1}{st}$$

Remark. *The A-module structure is achieved by "pulling back along the localization map" like so: Given $\psi : S^{-1}A \to \mathrm{End}_{\mathbb{Z}}(S^{-1}M)$, we take $f : A \to S^{-1}A$ and compose to get $\rho_A : A \to \mathrm{End}_{\mathbb{Z}}(S^{-1}M)$ where $\rho_A = \psi \circ f$*

Proof. Let M and $S^{-1}M$ be A-modules, with $S^{-1}M$ defined as above. Define the map of A-modules

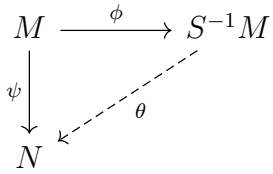
$$\phi : M \to S^{-1}M$$
$$\phi : m \mapsto \frac{m}{1}$$

We want to show that this map exists and that it satisfies the universal property. The category we are working in is the (full) subcategory of the slice category M, where objects are pairs (N, ψ) where $\psi : M \to N$ and every element of S maps to an invertible in N. Morphisms, say between (N_1, ψ_1) and (N_2, ψ_2) are maps $\theta : N_1 \to N_2$ which make



commute (so $\psi_2 = \theta \circ \psi_1$). So we want to show $\phi : M \to S^{-1}M$ is initial in this category (or, the pair $(S^{-1}M, \phi)$ is an initial object).

So, let (N, ψ) be an object of this category. We want to show there $\exists! \theta : S^{-1}M \to N$ such that



commutes ($\psi = \theta \circ \phi$).

Uniqueness of θ is seen by:

$$\begin{aligned} \theta\left(\frac{m}{s}\right) &= \theta\left(\mu_s^{-1}\left(\phi(m)\right)\right) \\ &= \mu_s^{-1}\left(\theta\left(\phi(m)\right)\right) \\ &= \mu_s^{-1}\left(\psi(m)\right) \end{aligned}$$

Where the second equality is due to $\theta\left(\mu_s(m)\right) = \mu_s\left(\theta(m)\right)$, then multiply by μ_s^{-1} (which exists) on both sides.

Well-defined: Let $\frac{m}{s} = \frac{m'}{s'}$, then $\exists t \in S$ with $t(ms' - m's) = 0$. Apply $\psi : M \to N$ to both sides:

$$t(\psi(m)s' - \psi(m')s) = 0$$

Since $t \in S$ is a unit, it is not a zerodivisor, so $s' \cdot \psi(m) = s \cdot \psi(m')$. Since $s, s' \in S$, $s \cdot s' \in S$ and so $\mu_{s \cdot s'}^{-1}$ exists. Multiply by this on both sides:

$$\mu_s^{-1}\psi(m) = \mu_{s'}^{-1}\psi(m') \implies \theta\left(\frac{m}{s}\right) = \theta\left(\frac{m'}{s'}\right)$$

θ respects addition:

$$\begin{aligned}
\theta\left(\frac{m_1}{s_1} + \frac{m_2}{s_2}\right) &= \theta\left(\frac{m_1 s_2 + m_2 s_1}{s_1 s_2}\right) \\
&= \mu_{s_1 s_2}^{-1}\left(\psi(m_1)\mu_{s_2} + \psi(m_2)\mu_{s_1}\right) \\
&= \mu_{s_1}^{-1}\left(\psi(m_1)\right) + \mu_{s_2}^{-1}(\psi(m_2)) \\
&= \theta\left(\frac{m_1}{s_1}\right) + \theta\left(\frac{m_2}{s_2}\right)
\end{aligned}$$

θ respects A-scalar multiplication:

$$\begin{aligned}
\theta\left(a \cdot \frac{m}{s}\right) &= \theta\left(\frac{a \cdot m}{s}\right) \\
&= \mu_s^{-1}\left(\psi(a \cdot m)\right) \\
&= \mu_s^{-1}\left(a \cdot \psi(m)\right) \\
&= a \cdot \mu_s^{-1}\left(\psi(m)\right) \\
&= a \cdot \theta\left(\frac{m}{s}\right)
\end{aligned}$$

Where the 4th equality is due to μ_s^{-1} being the inverse of an A-linear operation, so it is A-linear.

So θ exists and is unique, therefore $(S^{-1}M, \phi)$ is initial in this category. $\square$

1.2.F Some Localization Problems

(a) Show that localization commutes with finite products (equivalently, with finite direct sums).

(b) Show that localization commutes with arbitrary direct sums.

(c) Show that localization does not necessarily commute with infinite products: the obvisou map $S^{-1}(\prod_i M_i) \to \prod_i S^{-1}M_i$ induced by the universal property is not always an isomorphism.
(Hint: $(1, \frac{1}{2}, \frac{1}{3}, \dots) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \times \cdots$)

Proof. (a). We need to show there is a bijective A-module morphism

$$f : S^{-1}(M_1 \times \cdots \times M_n) \to S^{-1}M_1 \times \cdots \times S^{-1}M_n$$

Since $S^{-1}(M_1 \times \cdots \times M_n)$ is a localization, the map

$$\phi : M_1 \times \cdots \times M_n \to S^{-1}(M_1 \times \cdots \times M_n)$$

is initial among A-module maps out of $M_1 \times \cdots \times M_n$ where $s \in S$ acts invertibly. Consider the map

$$\psi : M_1 \times \cdots \times M_n \to S^{-1}M_1 \times \cdots \times S^{-1}M_n$$

$$\psi : (m_1, \dots, m_n) \mapsto (\psi_1(m_1), \dots, \psi_n(m_n))$$

Where $\psi_i : M_i \to S^{-1}M_i$ is the canonical localization map. This is an A-module homomorphism out of $M_1 \times \cdots \times M_n$. And since $\mu_s \in \mathrm{End}_{\mathbb{Z}}(S^{-1}M_1 \times \cdots \times S^{-1}M_n)$ acts componentwise, and s acts invertibly on each component $\phi_i(m_i)$, s acts invertibly on the product. So $\mu_s^{-1} \in \mathrm{End}_{\mathbb{Z}}(S^{-1}M_1 \times \cdots \times S^{-1}M_n)$. Therefore the pair $(S^{-1}M_1 \times \cdots \times S^{-1}M_n, \psi)$ is an object of this category. So by the universal property of localization:

$$\begin{array}{ccc} M_1 \times \cdots \times M_n & \xrightarrow{\quad\phi\quad} & S^{-1}(M_1 \times \cdots \times M_n) \\ \Big\downarrow{\scriptstyle \psi} & \swarrow{\scriptstyle f} & \\ S^{-1}M_1 \times \cdots \times S^{-1}M_n & & \end{array}$$

$$\exists! f : S^{-1}(M_1 \times \cdots \times M_n) \to S^{-1}M_1 \times \cdots \times S^{-1}M_n$$
$$f : \frac{(m_1, \ldots, m_n)}{s} \mapsto \left(\frac{m_1}{s}, \ldots, \frac{m_n}{s}\right)$$

Which makes the diagram commute. We want to show f is a bijective A-module homomorphism.

Surjective: Let $\left(\frac{m_1}{s_1}, \ldots, \frac{m_n}{s_n}\right) \in S^{-1}M_1 \times \cdots \times S^{-1}M_n$. Define $t = \prod_i s_i$, since S is multiplicative, $t \in S$. Let $x_i = \prod_{j \neq i} s_j \in S$, then

$$\left(\frac{m_1}{s_1}, \ldots, \frac{m_n}{s_n}\right) = \left(\frac{x_1 m_1}{t}, \ldots, \frac{x_n m_n}{t}\right) = \theta\left(\frac{(x_1 m_1, \ldots, x_n m_n)}{t}\right)$$

So θ is surjective.

Injective: Let $\theta\left(\frac{(m_1,\ldots,m_n)}{s}\right) = \theta\left(\frac{(m_1',\ldots,m_n')}{t}\right)$. So

$$\left(\frac{m_1}{s}, \ldots, \frac{m_n}{s}\right) = \left(\frac{m_1'}{t}, \ldots, \frac{m_n'}{t}\right)$$

Then componentwise, $\frac{m_i}{s} = \frac{m_i'}{t} \implies \exists u_i \in S$ such that

$$u_i(m_i t - m_i' s) = 0$$

Then let $u = \prod_i u_i$. Clearly $u \in S$, and

$$u(m_i t - m_i' s) = 0$$

For all $i \in I$ simultaneously. So

$$u\left[(m_1, \ldots, m_n)t - (m_1', \ldots, m_n')s\right] = 0$$

Therefore

$$\frac{(m_1, \ldots, m_n)}{s} = \frac{(m_1', \ldots, m_n')}{t}$$

As required.

Alternatively: Let $\theta\left(\frac{(m_1,\ldots,m_n)}{s}\right) = 0$. Then $\frac{m_i}{s} = 0$ for all i. Therefore

$$\frac{(m_1, \ldots, m_n)}{s} = 0$$

so $\ker(\theta) = \{0\}$, and so θ is injective.

(b). Let $\theta : S^{-1}\left(\bigoplus_{i\in I} m_i\right) \to \bigoplus_{i\in I} S^{-1}M_i$ be the unique A-module homomorphism given by the universal property of localization, which maps

$$\theta : \frac{(m_i)_{i\in I}}{s} \mapsto \left(\frac{m_i}{s}\right)_{i\in I}$$

It remains to show that θ is bijective.

Surjective: Let

$$x = (x_i)_{i\in I} \in \bigoplus_{i\in I} S^{-1}M_i$$

be an arbitrary element. Since it is an element of a direct sum, it has finite support. Let $J \subset I$ be the (finite) subset of I such that $x_j = \frac{m_j}{s_j} \neq 0$ when $j \in J$, and $x_i = 0$ otherwise. The collection of all such x_j terms defines an isomorphism into $S^{-1}M_1 \times \cdots \times S^{-1}M_n$. Then, we use the argument from (a) to see θ is surjective.

Injective: Let

$$\theta\left(\frac{(m_i)_{i\in I}}{s}\right) = 0$$

Then $\frac{m_i}{s} = 0$ for all $i \in I$. Let $J \subset I$ again be the finite subset for which $m_j \neq 0$, and $m_i = 0$ otherwise. If $j \in J$, then

$$m_j = 0 \implies \exists u_j \in S \text{ such that } u_j m_j = 0$$

Define $u = \prod_{j\in J} u_j$. Then for all i, $um_i = 0$. So we have found a $u \in S$ such that $um_i = 0$ for all $i \in I$, therefore $\frac{(m_i)_{i\in I}}{s} = 0$, and so $\ker(\theta) = \{0\}$ and θ is injective.

(c). Let $A = \mathbb{Z}$, $S = \mathbb{Z} \setminus \{0\}$. Then $S^{-1}\mathbb{Z} = \mathbb{Q}$. Consider the element

$$\left(1, \frac{1}{2}, \frac{1}{3}, \dots\right) \in \mathbb{Q} \times \mathbb{Q} \times \dots$$

A valid preimage of this element would have to contain every positive integer (in particular every prime number) as a factor of some $s \in S$. However that is a contradiction because no integer contains every prime as a factor. So

$$\theta : S^{-1}\left(\prod \mathbb{Z}\right) \to \prod \mathbb{Q}$$

Is not an isomorphism.

□

1.2.G Calculating a Tensor Product

Show that

$$\mathbb{Z}/(10) \otimes_{\mathbb{Z}} \mathbb{Z}/(12) \cong \mathbb{Z}/(2)$$

Remark. *Note: Over $\mathbb{Z}$, the tensor product $M \otimes_{\mathbb{Z}} N = \{m \otimes n \mid m \in M, n \in N\}$ is completely determined by the pure tensor $1 \otimes 1$, because for any $m \otimes n \in M \otimes_{\mathbb{Z}} N$, we have*

$$m \otimes n = m(1 \otimes n) = mn(1 \otimes 1)$$

Proof. Let $x \otimes y \in \mathbb{Z}/(10) \otimes_{\mathbb{Z}} \mathbb{Z}/(12)$. Then, $x \otimes y = xy(1 \otimes 1)$. By Bezout, since $\gcd(10, 12) = 2$, we can write $2 = 10(-1) + 12(1)$ So for even integers $2k$ we have

$$2k = [10(-1) + 12(1)]\, k$$

So

$$\begin{aligned} 2k(1 \otimes 1) &= [10(-1) + 12(1)]\, k(1 \otimes 1) \\ &= -10k(1 \otimes 1) + 12k(1 \otimes 1) \\ &= 0 + 0 = 0 \end{aligned}$$

For odd integers $2k + 1$ we have

$$2k + 1 = [10(-1) + 12(1)]\, k + 1$$

So

$$2k + 1(1 \otimes 1) = 2k(1 \otimes 1) + 1(1 \otimes 1) = 1 \otimes 1$$

So there are 2 cases:

- If $2 \mid xy$, then $xy(1 \otimes 1) = 0$
- If $2 \nmid xy$, then $xy(1 \otimes 1) = 1 \otimes 1$

(Furthermore, $(1 \otimes 1) + (1 \otimes 1) = 2(1 \otimes 1) = 0$)
Therefore, $\mathbb{Z}/(10) \otimes_{Z} \mathbb{Z}/(12) \cong \mathbb{Z}/(2)$. □

1.2.H Right Exactness of Tensor Product

Show that $(-) \otimes_A N$ is a covariant functor $\mathbf{Mod_A} \to \mathbf{Mod_A}$ and that it is right-exact. So if

$$0 \longrightarrow M' \xrightarrow{g} M \xrightarrow{f} M'' \longrightarrow 0$$

Is an exact sequence of A-modules, then

$$M' \otimes_A N \xrightarrow{\overline{g}} M \otimes_A N \xrightarrow{\overline{f}} M'' \otimes_A N \longrightarrow 0$$

Is also exact.

Proof. First we will show that

$$F \coloneqq (-) \otimes_A N : \mathbf{Mod_A} \to \mathbf{Mod_A}$$

is a covariant functor. F sends an object $M \in \mathrm{Obj}(\mathbf{Mod_A})$ to $M \otimes_A N \in \mathrm{Obj}(\mathbf{Mod_A})$. For a morphism of A-modules $\phi : M_1 \to M_2$, F sends ϕ to $\overline{\phi} : M_1 \otimes_A N \to M_2 \otimes_A N$, where

$$\overline{\phi}(m_1 \otimes n) = \phi(m_1) \otimes n$$

We check:

- F preserves identities:

  $$F(id_M)(m \otimes n) = id_M(m) \otimes n = m \otimes n$$

  so $F(id_M) = id_{F(M))}$.

- F respects composition:

  $F(\phi \circ \psi)(m \otimes n) = (\phi \circ \psi)(m) \otimes n = \phi(\psi(m)) \otimes n = F(\phi)(F(\psi)(m \otimes n))$

  So $F(\phi \circ \psi) = F(\phi) \circ F(\psi)$.

Therefore F is a (covariant) functor. Now we need to show that F is right-exact. So let

$$0 \longrightarrow M' \xrightarrow{g} M \xrightarrow{f} M'' \longrightarrow 0$$

Be an exact sequence of A-modules. That is, f is a surjection and g is an injection. We want:

$$F(M') \xrightarrow{\overline{g}} F(M) \xrightarrow{\overline{f}} F(M'') \longrightarrow 0$$

To be exact. That is, $\overline{g}$ surjects onto $\ker(\overline{f})$, and $\overline{f}$ is a surjection.

(i). $\overline{f}$ surjection: Let $m'' \otimes n \in M'' \otimes_A N$ be a pure tensor. Then since $m'' \in M''$, and f surjects onto M'', $\exists m \in M$ with $f(m) = m''$. Then $m'' \otimes n = f(m) \otimes n = \overline{f}(m \otimes n)$. So $\overline{f}$ is a surjection.

(ii). $\mathrm{Im}(\overline{g}) \subseteq \ker(\overline{f})$: Let $x \in \mathrm{Im}(\overline{g})$, so $x = g(m) \otimes n$. But $\mathrm{Im}(g) = \ker(f)$, so $g(m) \in ker(f)$. Then

$$\overline{f}(x) = f(g(m)) \otimes n = 0 \otimes n = 0$$

So $x \in \ker(\overline{f})$.

(iii). By (i) and first isomorphism theorem,

$$M \otimes_A N/\ker(\overline{f}) \cong M'' \otimes N$$

By (ii), the inclusion $\mathrm{Im}(\overline{g}) \hookrightarrow \ker(\overline{f})$ induces a surjection

$$M \otimes_A N/\mathrm{Im}(\overline{g}) \twoheadrightarrow M \otimes_A N/\ker(\overline{f})$$

Composing gives a map

$$\hat{f} : M \otimes_A N/\mathrm{Im}(\overline{g}) \to M'' \otimes_A N$$

(iv). $h : M'' \times N \to M \otimes_A N/\mathrm{Im}(\overline{g})$ well defined:
Define $h(m'', n) = m \otimes n + \mathrm{Im}(\overline{g})$, where m is the preimage of m'' in f. Let $(m'', n) \in M'' \times N$, and choose m_1, m_2 such that $f(m_1 = m'')$ and $f(m_2) = m''$. Then

$$[m_1 \otimes n + \mathrm{Im}(\overline{g})] - [m \otimes n + \mathrm{Im}(\overline{g})] = (m_1 - m_2) \otimes n + \mathrm{Im}(\overline{g})$$

But $f(m_1 - m_2) = 0$, so $m_1 - m_2 \in \ker(f) \supseteq \mathrm{Im}(g)$. So $m_1 - m_2 \in \mathrm{Im}(g)$. Therefore $(m_1 - m_2) \otimes n + \mathrm{Im}(\overline{g}) = 0 + \mathrm{Im}(\overline{g})$, so h is well defined.

(v). h is A-bilinear :

$$\begin{aligned} h(m_1'' + m_2'', n) &= (m_1 + m_2) \otimes n + \operatorname{Im}(\overline{g}) \\ &= [m_1 \otimes n + \operatorname{Im}(\overline{g})] + [m_2 \otimes n + \operatorname{Im}(\overline{g})] \\ &= h(m_1'', n) + h(m_2'', n) \end{aligned}$$

So h is linear in the first term.

$$\begin{aligned} h(m'', n_1 + n_2) &= m \otimes (n_1 + n_2) + \operatorname{Im}(\overline{g}) \\ &= [m \otimes n_1 + \operatorname{Im}(\overline{g})] + [m \otimes n_2 + \operatorname{Im}(\overline{g})] \\ &= h(m'', n_1) + h(m'', n_2) \end{aligned}$$

So h is linear in the second term.

$$h(am'', n) = (am) \otimes n + \operatorname{Im}(\overline{g}) = a(m \otimes n) + \operatorname{Im}(\overline{g}) = a \cdot h(m'', n)$$

And similar for $h(m'', an)$. So h is A-bilinear.
Therefore, by the universaal property of tensor products:

$$\begin{array}{ccc} M'' \times N & \xrightarrow{\quad\pi\quad} & M'' \otimes_A N \\ \Big\downarrow{\scriptstyle h} & \swarrow{\scriptstyle \exists! \psi} & \\ M \otimes_A N / \operatorname{Im}(\overline{g}) & & \end{array}$$

Where $h = \psi \circ \pi$.

Finally, we need to show that ψ and $\hat{f}$ are inverses. We already know $\hat{f}$ is surjective, so it suffices to show $\psi \circ \hat{f} = id_{M \otimes_A N / \operatorname{Im}(\overline{g})}$, which implies that $\hat{f}$ is injective. We check:

$$\psi(\hat{f}(m \otimes n + \operatorname{Im}(\overline{g}))) = \psi(f(m) \otimes n) = m \otimes n + \operatorname{Im}(\overline{g})$$

So $\hat{f}$ is injective, and therefore bijective. Therefore,

$$M \otimes_A N / \operatorname{Im}(\overline{g}) \cong M \otimes_A N / \ker(\overline{f})$$

So $\operatorname{Im}(\overline{g}) = \ker(\overline{f})$ and the sequence

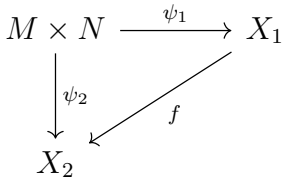
$$M' \otimes_A N \xrightarrow{\ \overline{g}\ } M \otimes_A N \xrightarrow{\ \overline{f}\ } M'' \otimes_A N \longrightarrow 0$$

Is exact, so the functor $(-) \otimes_A N : Mod_A \to Mod_A$ is right exact. $\square$

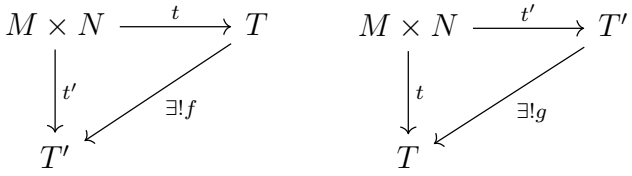
1.2.I Uniqueness Of Universal Pair

Show that (t, T), where $t : M \times N \to T$ is bilinear in A, is unique up to unique isomorphism.

Proof. The category we are working in is the subcategory of the coslice category under $M \times N$ such that morphisms $M \times N \to X$ are A-bilinear. That is, objects are pairs (ψ, X) where $\psi : M \times N \to X$ is A-bilinear. Morphisms are the diagonal maps:



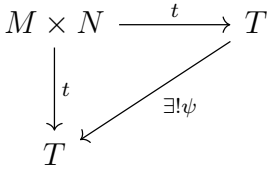
such that $\psi_2 = f \circ \psi_1$.
Let (t, T) and (t', T') be two objects both satisfying the universal property, so



Both commute. By the universal property for (t, T), $\exists! f : T \to T'$ such that $t' = f \circ t$. Similarly by the universal property for (t', T'), $\exists! g : T' \to T$ such that $t = g \circ t'$. Then

$$t = g \circ t' = g \circ (f \circ t) = (g \circ f) \circ t$$

and vice versa. Therefore $g \circ f$ is a morhism $T \to T$ satisfying:



Where $\psi \circ t = t$, so $g \circ f = id_T$. Similarly, $f \circ g = id_{T'}$. Therefore (t, T) is unique up to unique isomorphism. □

1.2.J Univseral Property of Tensor Products

Show that the construction of 1.2.5 satisfies the universal property of tensor products.

Proof. Let $F(M \times N)$ be the free module generated by $M \times N$. Let R be the submodule generated by the following relations:

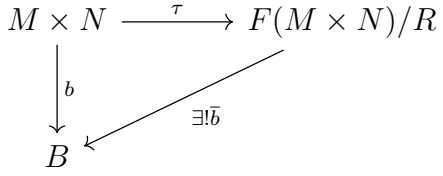
- $a(m,n) - (am,n)$
- $a(m,n) - (m,an)$
- $(m_1 + m_2, n) - (m_1,n) - (m_2,n)$
- $(m, n_1 + n_2) - (m,n_1) - (m,n_2)$

Then the quotient module $F(M\times N)/R$ consists of finite linear combinations of pairs (m,n) subject to those relations. Let

$$\tau : M \times N \to F(M \times N)/R$$

$$\tau(m,n) = \overline{(m,n)}$$

be the canonical map. We want to show that the pair $\tau, F(M\times N)/R$ is initial among A-bilinear maps out of $M \times N$. In other words, let $b : M \times N \to B$ be A-bilinear, then



There is a unique $\bar{b} : F(M \times N)/R \to B$ such that $b = \bar{b} \circ \tau$.

(i). Uniqueness: Let $f : F(M \times N)/R \to B$ be an A-linear map with $\overline{b = f \circ \tau}$. Then for any pair (m,n) we have

$$f(\tau((m,n))) = f(\overline{(m,n)}) = b(m,n)$$

Any element of $F(M \times N)/R$ is a finite linear combination of these pairs. Since f is A-linear and determined by b, f is unique.

(ii). existence: We must show $\overline{b}$ is well defined on $F(M \times N)/R$, and that $\overline{b}$ is a valid A-module homomorphism.

(a). Well defined: Let $\overline{x} = 0$ in $F(M \times N)/R$. By definition, $x \in R$. Since $x \in R$, it looks like a finite linear combination of the relations defined above. So then $\overline{b}(\overline{x}) = b(x)$, since b is an A-bilinear map, we have that all the relations defining R hold. Therefore, $b(x) = 0$ and so $\overline{b}$ is well defined.

(b). $\overline{b}$ is an A-module homomorphism: Let $x, y \in F(M \times N)/R$, and let $a \in A$.

$$\begin{aligned} \overline{b}\left(\overline{x} + \overline{y}\right) &= \overline{b}\left(\overline{x + y}\right) \\ &= b\left(x + y\right) \\ &= b\left(x\right) + b\left(y\right) \\ &= \overline{b}(\overline{x}) + \overline{b}(\overline{y}) \end{aligned}$$

and

$$\begin{aligned} a\overline{b}(\overline{x}) &= ab(x) \\ &= b(ax) \\ &= \overline{b}(\overline{ax}) \\ &= \overline{b}(a\overline{x}) \end{aligned}$$

So $\overline{b}$ is a valid A-module homomorphism.

Therefore, $(\tau, F(M \times N)/R)$ is initial among A-bilinear maps out of $M \times N$. □

1.2.K Extension of Scalars

(a). Let A be a ring, let M be an A-module, and let $f : A \to B$ be a ring homomorphism. We know $B \otimes_A M$ is an A-module. Show that it is also a B-module, and that this defines a functor from $Mod_A \to Mod_B$.

Proof. We need a function $\phi : B \times (B \otimes_A M) \to B \otimes_A M$ which satisfies the B-module axioms. We will do so by constructing maps $B \otimes_A M \to B \otimes_A M$ for each $b \in B$. Fix $b \in B$ and consider the map

$$\psi_b : B \times M \to B \otimes_A M$$

$$\psi_b(b', m) = bb' \otimes m$$

(i). Linearity in the first term:

$$\begin{aligned}\psi_b(b'_1 + b'_2, m) &= b(b'_1 + b'_2) \otimes m \\ &= bb'_1 \otimes m + bb'_2 \otimes m \\ &= \psi_b(b'_1, m) + \psi_b(b'_2, m)\end{aligned}$$

(ii). Linearity in the second term:

$$\begin{aligned}\psi_b(b', m_1 + m_2) &= bb' \otimes (m_1 + m_2) \\ &= bb' \otimes m_1 + bb' \otimes m_2 \\ &= \psi_b(b', m_1) + \psi_b(b', m_2)\end{aligned}$$

(iii). Scalar multiplication:

$$\begin{aligned}\psi_b(ab', m) &= bab' \otimes m \\ &= abb' \otimes m \qquad \text{(Rings commutative by convention)} \\ &= bb' \otimes am \\ &= a(bb' \otimes m) \\ &= a \cdot \psi_b(b', m)\end{aligned}$$

So ψ_b is A-bilinear for all $b \in B$. By the universal property of tensor products, each ψ_b factors uniquely through $B \otimes_A M$. So for each $b \in B$, there is an induced map:

$$\phi_b : B \otimes_A M \to B \otimes_A M$$

$$\phi_b(b' \otimes m) = bb' \otimes m$$

So using these ϕ_b, we can define

$$\phi : B \times (B \otimes_A M) \to B \otimes_A M$$

$$\phi(b, b' \otimes m) = \phi_b(b' \otimes m)$$

Then we check:

(i). Identity:

$$\begin{aligned}\phi(1_B, b' \otimes m) &= \phi_{1_B}(b' \otimes m) \\ &= (1_B b') \otimes m \\ &= b' \otimes m\end{aligned}$$

(ii). Linearity in first term:

$$\begin{aligned}\phi(b_1 + b_2, b' \otimes m) &= \phi_{b_1+b_2}(b' \otimes m) \\ &= [(b_1 + b_2)b'] \otimes m \\ &= b_1 b' \otimes m + b_2 b' \otimes m \\ &= \phi(b_1, b' \otimes m) + \phi(b_2, b' \otimes m)\end{aligned}$$

(iii). Linearity in second term:

$$\begin{aligned}\phi(b, b'_1 \otimes m_1 + b'_2 \otimes m_2) &= \phi_b(b'_1 \otimes m + b'_2 \otimes m) \\ &= \phi_b(b'_1 \otimes m) + \phi_b(b'_2 \otimes m) \\ &= \phi(b, b'_1 \otimes m) + \phi(b, b'_2 \otimes m)\end{aligned}$$

(iv). B-scalar multiplication:

$$\begin{aligned}\phi(b_1 b_2, b' \otimes m) &= \phi_{b_1 b_2}(b' \otimes m) \\ &= (b_1 b_2)b' \otimes m \\ &= b_1(b_2 b' \otimes m) \\ &= b_1(\phi_{b_2}(b' \otimes m)) \\ &= \phi_{b_1}(\phi_{b_2}(b' \otimes m)) \\ &= \phi(b_1, \phi(b_2, b' \otimes m))\end{aligned}$$

So ϕ is a valid B-module map, so $B \otimes_A M$ is a B-module. Therefore, for any A-module M (any object of $\mathbf{Mod_A}$) we can construct a functor

$$F : \mathbf{Mod_A} \to \mathbf{Mod_B}$$

So that $F(M) \coloneqq B \otimes_A M$ is the object assignment, and for an A-module morphism $g : M \to N$, we assign a map $F(g) = \overline{g} : B \otimes_A M \to B \otimes_A N$. We need to show that $\overline{g}$ exists and is unique.
So let $g : M \to N$ be an A-module morphism. Define the map:

$$\hat{g} : B \times M \to B \otimes_A N$$

$$\hat{g}(b, m) = b \otimes g(m)$$

We claim $\hat{g}$ is A-bilinear. We check:

(i). Linearity in the first term

$$\begin{aligned} \hat{g}(b_1 + b_2, m) &= b_1 + b_2 \otimes g(m) \\ &= b_1 \otimes g(m) + b_2 \otimes g(m) \\ &= \hat{g}(b_1, m) + \hat{g}(b_2, m) \end{aligned}$$

(ii). Linearity in the second term

$$\begin{aligned} \hat{g}(b, m_1 + m_2) &= b \otimes g(m_1 + m_2) \\ &= b \otimes g(m_1) + g(m_2) \qquad \text{(By linearity of } g\text{)} \\ &= b \otimes g(m_1) + b \otimes g(m_2) \\ &= \hat{g}(b, m_1) + \hat{g}(b, m_2) \end{aligned}$$

(iii). A-scalar multiplication:

$$\begin{aligned} a\hat{g}(b, m) &= a \cdot (b \otimes g(m)) \\ &= ab \otimes g(m) \\ &= \hat{g}(ab, m) \\ &= b \otimes ag(m) \\ &= b \otimes g(am) \\ &= \hat{g}(b, am) \end{aligned}$$

Therefore $\hat{g}$ is an A-bilinear map out of $B \times M$. By the universal property of tensor products, there is a unique A-module morphism

$\overline{g} : B \otimes_A M \to B \otimes_A N$. We need to show that $\overline{g}$ is actually a B-module morphism. We check:

$$\begin{aligned}\overline{g}(b \cdot b' \otimes m) &= bb' \otimes g(m) \\ &= b(b' \otimes g(m)) \\ &= b\overline{g}(b' \otimes m)\end{aligned}$$

and since $\overline{g}$ is an A-module morphism, we have $\overline{g}(x+y) = \overline{g}(x) + \overline{g}(y)$. So $\overline{g}$ is a B-module morphism. So the functor F maps:

$$F : \mathbf{Mod_A} \to \mathbf{Mod_B}$$

$$F(M) = B \otimes_A M, \qquad F(g) = \overline{g}$$

Finally, we need to check that F is a functor:

(i). Identity: $F(id_M) = \overline{id_M}$, which is

$$\overline{id_M}(b \otimes m) = b \otimes id_M(m) = b \otimes m$$

So $F(id_M) = id_{B \otimes_A M} = id_{F(M)}$

(ii). Composition: $F(f \circ g) = \overline{f \circ g}$ and $F(f) \circ F(g) = \overline{f} \circ \overline{g}$.

$$(\overline{f \circ g})(b \otimes m) = b \otimes (f \circ g)(m) = b \otimes f(g(m))$$

$$(\overline{f} \circ \overline{g})(b \otimes m) = \overline{f}(b \otimes g(m)) = b \otimes f(g(m))$$

So $F(f \circ g) = F(f) \circ F(g)$

Therefore F is a functor. □

(b). If $A \to C$ is another morphism of rings, show that $B \otimes_A C$ has a natural structure of a ring. Hint: Multiplication will be given by

$$(b_1 \otimes c_1)(b_2 \otimes c_2) = (b_1 b_2) \otimes (c_1 c_2)$$

Proof. Since $h : A \to C$ is a ring homomorphism, C has the structure of an A-module by $a \cdot b = h(a)b$. Therefore by (a), we know $B \otimes_A C$ has the structure of an A-module. Therefore $B \otimes_A C$ is an abelian group. It remains to show that there is a multiplication on $B \otimes_A C$ which satisfies the ring axioms:

(i). $\exists 1_{B\otimes_A C} \in B \otimes_A C$ with $1_{B\otimes_A C} \cdot x = x \cdot 1_{B\otimes_A C} = x$

(ii). $x \cdot (y + z) = x \cdot y + x \cdot z$ for $x, y, z \in B \otimes_A C$

(iii). $x \cdot (y \cdot z) = (x \cdot y) \cdot z$ for $x, y, z \in B \otimes_A C$

Let

$$\cdot : B \otimes_A C \times B \otimes_A C \to B \otimes_A C$$

$$\cdot : [(b_1 \otimes c_1), (b_2 \otimes c_2)] \mapsto (b_1 b_2) \otimes (c_1 c_2)$$

Using the multiplication in B and C, respectively.
Proof of (i): Since B and C are rings, by our convention $\exists 1_B \in B$ and $1_C \in C$. Let $1_B \otimes 1_C \in B \otimes_A C$. We check:

$$(1_B \otimes 1_C) \cdot (b \otimes c) = (1_B b) \otimes (1_C c) = b \otimes c$$

So $1_B \otimes 1_C$ is the identity in $B \otimes_A C$.
Proof of (ii): Let $b_1, b_2, b_3 \in B$ and $c_1, c_2, c_3 \in C$.

$$\begin{aligned}(b_1 \otimes c_1) \cdot [b_2 \otimes c_2 + b_3 \otimes c_3] &= (b_1 \otimes c_1) \cdot (b_2 + b_3 \otimes c_2 + c_3) \\ &= (b_1(b_2 + b_3)) \otimes c_1(c_2 + c_3) \\ &= b_1 b_2 + b_1 b_3 \otimes c_1 c_2 + c_1 c_3\end{aligned}$$

And

$$\begin{aligned}b_1 \otimes c_1 \cdot b_2 \otimes c_2 + b_1 \otimes c_1 \cdot b_3 \otimes c_3 &= b_1 b_2 \otimes c_1 c_2 + b_1 b_3 \otimes c_1 c_3 \\ &= b_1 b_2 + b_1 b_3 \otimes c_1 c_2 + c_1 c_3 \\ &= (b_1 \otimes c_1) \cdot [b_2 \otimes c_2 + b_3 \otimes c_3]\end{aligned}$$

So multiplication distributes over $+$.
Proof of (iii): Let $b_1, b_2, b_3 \in B$ and $c_1, c_2, c_3 \in C$.

$$\begin{aligned}(b_1 \otimes c_1) \cdot [(b_2 \otimes c_2) \cdot (b_3 \otimes c_3)] &= (b_1 \times c_1) \cdot (b_2 b_3 \otimes c_2 c_3) \\ &= b_1(b_2 b_3) \otimes c_1(c_2 c_3) \\ &= (b_1 b_2) b_3 \otimes (c_1 c_2) c_3 \\ &= [(b_1 \otimes c_1) \cdot (b_2 \otimes c_2)] \cdot (b_3 \otimes c_3)\end{aligned}$$

So $\cdot$ is associative. Therefore, $B \otimes_A C$ has the strcture of a ring. $\square$

1.2.L Tensoring Localization with Module is Localized Module

If S is a multiplicative subset of A and M is an A-module, describe a natural isomorphism

$$(S^{-1}A) \otimes_A M \xrightarrow{\sim} S^{-1}M$$

As A-modules and as $S^{-1}A$-modules.

Proof. Let

$$f : (S^{-1}A) \times M \to S^{-1}M$$
$$f : \left(\frac{a}{s}, m\right) \mapsto \frac{a \cdot m}{s}$$

□